1.2 Applications of Propositional Logic

Tuesday, February 15, 2022 5:59 PM

✓ 1. $a \to e$

2. $(e \vee p) \to m$

✗ 3. $(m \wedge r \wedge b) \to g$ $g \to (r \wedge (\neg m) \wedge (\neg b))$

4. $(d \vee s) \to w$

✗ 5. $a \wedge (b \vee (p \wedge r)) \to e$ $e \to (a \wedge (b \vee p) \wedge r)$

6. $(b_{32} \wedge g_1 \wedge r_1 \wedge h_{16}) \vee (b_{64} \wedge g_2 \wedge r_2 \wedge b_{32}) \to u$

7.

✓ a) $q \to p$

✗ b) $\neg p \to q$ Since "but" means "and", we have $q \wedge \neg p$

✗ c) $p \leftrightarrow q$ The sentence is saying the same thing as the sentence in part(a), so the answer is $q \to p$

✗ d) $q \to \neg p$ Again we recall that "when" means "if" in logic: $\neg q \to \neg p$

8.

a) $r \wedge \neg p$

b) $p \wedge r$

c) $\neg q \to \neg r$

d) $q \to (\neg p \wedge r)$

✓ 9. a = The system is in a multi-user state

b = It is operating normally

c = The Kernel is functioning

d = The system is in interupt mode

$a \leftrightarrow b$ $b \to c$ $\neg c \vee d$ $\neg a \to d$ $\neg d$

These system specifications are not consistent. It is stated that the system is not in interupt mode but in order for the system to be operating normally, it can only be in multiuser mode. This is a contradiction to the statement $\neg a \to d$ because if this condition is true, then the statement $\neg d$ is false.

10. a = The software system is being upgraded

b = users cannot access the file system

c = users can save new files

$a \to b$ $\neg b \to c$ $\neg c \to \neg a$

The system is not consistent. If software is being updated then users not having access contradicts if users cannot save files then the software is being updated because users need access to the file system to save new files.